

1.1 Introduction to ANOVA

$$H_0: \mu_1 = \mu_2 = \dots = \mu_k$$

$$H_1: \text{Some } \mu_k \neq \mu_j$$

$$Y = \mu_k + \epsilon$$

Total variation = Variation in response y explained by model + Unexplained variation in residuals

$$(y - \bar{y}) = (\bar{y}_k - \bar{y}) + (y - \bar{y}_k)$$

$$\sum (y - \bar{y})^2 = \sum (\bar{y}_k - \bar{y})^2 + \sum (y - \bar{y}_k)^2$$

$$SS_{\text{Total}} = SS_{\text{groups}} + SSE$$

Source	df	S.S.	M.S.	t.s.	p-value
Groups	$k-1$	SS_{groups}	$\frac{SS_{\text{groups}}}{k-1}$	$\frac{MS_{\text{groups}}}{MSE}$	use $F_{k-1, n-k}$
Error	$n-k$	SSE	$\frac{SSE}{n-k}$	MSE	
Total	$n-1$	SS_{Total}			

$$Y = \underbrace{\mu}_{\text{Grand Mean}} + \underbrace{\alpha_k}_{\text{Effect for } k^{\text{th}} \text{ group}} + \underbrace{\epsilon}_{\text{random error}}$$

$$\mu_k = \mu + \alpha_k \quad \hat{\alpha}_k = \bar{y}_k - \bar{y}$$

$$H_0: \alpha_1 = \alpha_2 = \dots = \alpha_k = 0$$

$$H_1: \text{Some } \alpha_k \neq 0$$

$$\epsilon \sim N(0, \sigma_\epsilon)$$

Zero Mean Always holds for sample residuals

Constant Variance Plot residuals v.s. fits and/or compare std. dev's of groups **Note** ^{also} crucial if sample sizes are equal.

Normality Histogram/normal plot of residuals

Independence Pay attention to data collection.

1.2 ANOVA Inference

Allocation of Units to Groups
Not

Using Randomisation

Using Randomisation

Selection of units
not

At Random

Random sample selected; units assigned randomly to treatment groups (I)

Random samples selected from separate populations (II)

← Inference can be drawn to populations

At Random

Study units are found, then randomly assigned to treatment groups (III)

Available units from separate populations are studied (IV)

↑
Causal inferences can be drawn

Fisher's LSD : pairwise t-tests

Single Mean

$$\bar{y}_i \pm t \sqrt{\frac{JMSE}{n_i}}$$

Group Means

$$\bar{y}_i - \bar{y}_j \pm t \sqrt{JMSE \left(\frac{1}{n_i} + \frac{1}{n_j} \right)}$$

2.1 Two-Way ANOVA

$$y = \mu + \alpha_k + \beta_j + \epsilon$$

$$SSTotal = SSA + SSB + SSE$$

$$\sum (y - \bar{y})^2 = \sum J(\bar{y}_k - \bar{y})^2 + \sum j(\bar{y}_j - \bar{y})^2 + (SSTotal - SSA - SSB)$$

Testing two hypotheses:

$$H_0: \alpha_1 = \alpha_2 = \dots = \alpha_k = 0$$

$$H_1: \text{Some } \alpha_k \neq 0$$

$$H_0: \beta_1 = \beta_2 = \dots = \beta_j = 0$$

$$H_1: \text{Some } \beta_j \neq 0$$

Simple Block Design

A simple block design has two factors with exactly one data value (observation) in each combination of the factors.

Interaction Effect

An interaction effect occurs when a significant difference is present at a specific combination of factors

$$y = \mu + \alpha_k + \beta_j + \gamma_{kj} + \epsilon$$

$$SSTotal = SSA + SSB + SSAB + SSE$$

$$\sum (y - \bar{y})^2 = \sum_k J(\bar{y}_k - \bar{y})^2 + \sum_j K(\bar{y}_j - \bar{y})^2 + \sum_{k,j} c(\bar{y}_{kj} - \bar{y}_k - \bar{y}_j + \bar{y})^2 + \sum (y - \bar{y}_{kj})^2$$

$$H_0: \alpha_1 = \alpha_2 = \dots = \alpha_k = 0$$

$$H_0: \beta_1 = \beta_2 = \dots = \beta_j = 0$$

$$H_0: \text{All } \gamma_{kj} = 0$$

Factorial Design

Levene's Test

$$H_0: \sigma_1^2 = \sigma_2^2 = \dots = \sigma_k^2$$

$$H_1: \text{Some } \sigma_i^2 \neq \sigma_j^2$$

Problem of Multiplicity

When doing many pairwise comparisons $\rightarrow$ likely to make a Type I error (find a false difference)
 $\rightarrow$ Fisher's LSD may be too lenient

Bonferroni Adjustment

When doing m tests with an overall error rate of α^* , use $\alpha \rightarrow \alpha^*/m$ for each test

Tukey's HSD

Comparison

A comparison looks at just two of the K possible groups in a study

$$\bar{y}_1 - \bar{y}_2, \bar{y}_1 - \bar{y}_3, \bar{y}_3 - \bar{y}_4$$

Contrast

A contrast looks at linear combinations of two or more of the K possible groups in a study, with the coefficients adding to zero.

$$\bar{y}_1 - \frac{1}{2}\bar{y}_2 - \frac{1}{2}\bar{y}_3$$

2.3 Non-parametric Tests

Wilcoxon-Mann-Whitney Test

Nonparametric parallel to two sample t-test

Conditions for the error terms:

Have median 0

Follow a continuous distribution

Be independent

$y = \theta_1$ for individuals in group 1

$y = \theta_2$ for individuals in group 2

Where θ_1 and θ_2 are population medians

$$H_0: \theta_1 = \theta_2$$

Idea: the normality condition is not met

Based on ranks - rank all the observations in both groups from smallest to largest

If null hypothesis is true, ranks for both groups can be considered to be randomly drawn from a box $[1, 2, \dots, m+n]$, so the sum of the ranks from one group should follow a dist.

Kruskal-Wallis Test

Nonparametric parallel to one-way ANOVA

Conditions for the error terms:

Have median 0

Follow a continuous dist.

Be independent

$$y = \theta_k \text{ for individuals in group } k$$

where θ_k is population median in group k

$$H_0: \theta_1 = \theta_2 = \dots = \theta_k$$

Friedman Test

Nonparametric parallel to two-way ANOVA for a simple block design

Conditions for the error terms:

Have median 0

Follow a continuous dist.

Be independent

$$y = \theta_k \text{ for individuals in group } k$$

where θ_k is population median in group k

$$H_0: \theta_1 = \theta_2 = \dots = \theta_k$$

Idea: ranking each row together, then considering the values of ranks by columns

2.2 ANCOVA

Covariate (nuisance variable)

We are interested in the response variable Y and the factor A

But a continuous variable X not of direct interest may have an effect on the Y - A relation

Covariate may affect may affect our conclusion about Y - A relation:

Significant relation Y - A may become insignificant after introducing X

Insignificant relation Y - A may become significant after introducing X

Basic idea: Use dummy predictors to convert an ANOVA for means into a regression model, and include covariance

$$\text{Active} = \beta_0 + \beta_1 \text{Rest} + \beta_2 \text{Male} + \beta_3 \text{Exer1} + \beta_4 \text{Exer2} + \epsilon$$

$\uparrow$ $\uparrow$ $\nwarrow$ $\uparrow$
Covariate Factor A Factor B

3.1 Experimental Studies

"Design": methods, strategies used to produce data together with the ways your choice of methods for data production. Design will shape the conclusions you can and cannot make based on your data.

Comparison is Needed

Obvious that, to draw a conclusion that a treatment is effective, one must compare treatment group and control (placebo, no treatment) group.

Historical Controls aren't enough: randomisation is needed

Experimenting

Randomise assignment of treatments to units:

Protect against bias:

- bias is due to systematic variation that makes the groups you compare different even before you apply the treatment
- if you can't randomise the assignment of subjects or experimental units to the treatment groups, you have no guarantee that the groups are comparable

Allow for causal inference

The logic of statistical testing relies on a process of elimination

If you randomise and group means are different, then there are only two possible reasons:

Either the observed diff are due to the
Or else they are due to the randomisation

If the p -value is tiny, then it is almost impossible to get the observed difference just from the randomisation $\therefore$ the treatments must be responsible for at least part of the observed difference

Justify using a probability model

Randomised Comparative Experiment

In a Randomised Comparative Experiment:

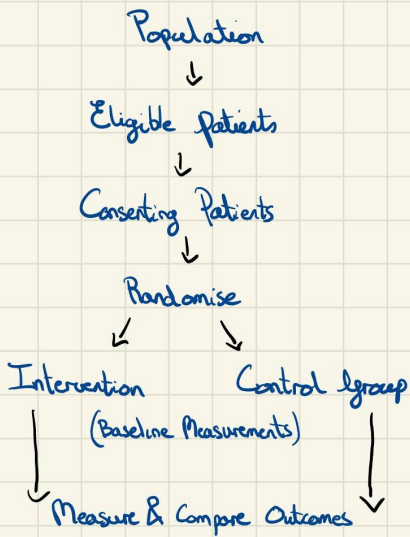
the experimenter must be able to assign the conditions to be compared

their assignable conditions are called treatments

the subjects, animals, or objects that receive the treatments are called experimental units

the treatments are assigned to units using chance

Conducting a Randomised Controlled Trial:



Look at:

Non-compliance

Completeness of follow-up

Random Allocation

Each person has the same probability (as the next person) of being allocated to each of the comparison groups. Usually, there is an equal probability of being allocated to each group

If allocation was truly random and the trial is reasonably large, factors which might affect outcome (e.g. age, past medical history) will be similar in each arm of the trial

∴ the estimate of effectiveness of the intervention is unbiased

Bias in Epidemiological Studies

Epidemiological studies estimate the prevalence of an exposure/outcome (cross-sectional study) or the association between an exposure and outcome (analytical studies e.g. RCT)

Bias relates to a departure from the true value

In general, bias can be classified as relating to:

- the selection of participants into (or out of) a study
- the measurement of exposure and/or outcome

Selection bias in RCTs

If bias in selection occurs equally to groups (e.g. only volunteers recruited) the estimate of association will be unbiased although generalisability may be affected.

Selection bias occurs when individuals allocated to the treatment and control groups differ in some characteristic associated with outcome

e.g. in a trial for depression, the recruiter may allocate those with more severe depression to the treatment and those less severe to the control group

Selection bias can result in an under/overestimate of the association between exposure and outcome

Loss to follow-up bias in RCT

If losses to follow-up (and reasons) occur equally across groups, the estimate of association will be unbiased although its precision may be reduced.

Loss to follow-up bias occurs when there is differential loss of participants between the groups and losses are related to outcome

e.g. treatment leads to side-effect that is more likely amongst older participants who are also more likely to have a worse outcome, if those experiencing the side-effect withdraw treatment effect will be overestimated

Loss to follow-up bias can result in an under/overestimate of the association between exposure and outcome

Measurement bias in RCT

Non-differential misclassification

- whether errors are random or systematic, if occur equally across treatment groups estimate of association will be underestimated

Differential misclassification

- if occur differently across treatment groups the association may be under or overestimated

Randomisation

Generation of unbiased allocation sequence

Concealment of allocation sequence

Sealed opaque envelopes

Central randomisation

Telephone randomisation

Avoids selection bias

Blinding

Being unaware of assigned intervention

Different parties that can be blinded

Participants in trial

Healthcare providers

} Avoid performance bias

Outcome assessors

Avoid detection bias

Often not feasible to blind all parties

Allocation concealment and blinding

These are not the same thing!

Allocation concealment means that those recruiting individuals are unaware of the random allocation schedule - this helps prevent selection bias

Blinding refers to subject/health care providers/outcome assessors being unaware of treatment allocation after randomisation has occurred - this helps prevent measurement bias

It is possible to achieve one without the other

'Intention to treat' analysis

Patients are analysed in the groups to which they were randomly allocated NOT on the basis of whether treatment was completed.

i.e. if a person stopped taking drug A after two days they would still be included in analysis in the drug A group

Avoids loss to follow-up bias through differential drop-out

The point of randomisation is the only time the two groups are truly comparable

i.e. ITT is the only unbiased estimate of effectiveness

Equipose

Do you 'know' treatment A is better than treatment B?

If you are uncertain whether A is better than B you can be described as being in 'equipose'

If you know A is better than B it would be unethical to give the patient a 50% chance of receiving poorer therapy

If you are in equipose then it would be ethical to seek informed consent

How do you 'know' A is better than B?

Why do we need observational studies?

Because RCTs may be:

unethical (e.g. exposure to radioactivity)

difficult to implement (e.g. income supplement)

inappropriate (e.g. rare outcome)

Observational studies may generate hypotheses that can be tested with an RCT

Randomisation F-Test

When assumptions of ANOVA are not met:

try and transform response

non parametric test

Randomisation F-test: empirical or data-driven approach to finding a p-value
permutation test

Steps:

- 1) Write each y -value on a card
- 2) Shuffle, then draw cards for the first group, for the second group, etc
- 3) Calculate a statistic (e.g. an ANOVA F-ratio) and record whether or not it exceeds the observed (real) value of that stat
- 4) Repeat
Report the fraction of extimes the observed is exceeded

- If there is no group diff (no treatment effects), then the only thing that can make one group diff from the other ^{is random assignment}
- If that's true, then the data I got should look much like other datasets I might have gotten by random assignment of response values to groups
- On the other hand, if my actual dataset looks quite diff from the typical datasets that I get by re-randomising then the diff is evidence that it is the treatments, not just the random assignments, causing the diff to be big

3.2 Observational Studies

Blocking

Block: A group of similar units, "similar" means similar with respect to the response variable

Reuse - repeated measures (e.g. before/after)

Subdivide (e.g. a plot of land)

Group (e.g. by age, weight, BP, ...)

Idea: to reduce variability within subjects, animals, or objects, apply treatments to similar units.

Repeated Measures Block Design (Within-Subject Design)

Each subject gets all the treatments, a different treatment in each of several time slots

Each treatment is given once to each subject

The order of the treatments is decided using chance, with a separate randomisation for each subject

Experimental unit: a time slot; each subject is a block of time slots.

Why creating blocks by reusing subjects may not make sense

Reusing subjects may not work because the treatment or measurement ↪ changes the subject
↪ takes too long

Blocks by Subdividing

Divide each block of land into equal-sized smaller plots, with the same number of plots per block, equal to the number of treatments

Randomly assign a treatment to each plot in such a way that each block gets all the treatments, one per plot, and each treatment goes to some plot in every block

Experimental unit: a plot; A block is a group of plots

Blocks by grouping (Matched Subjects Design)

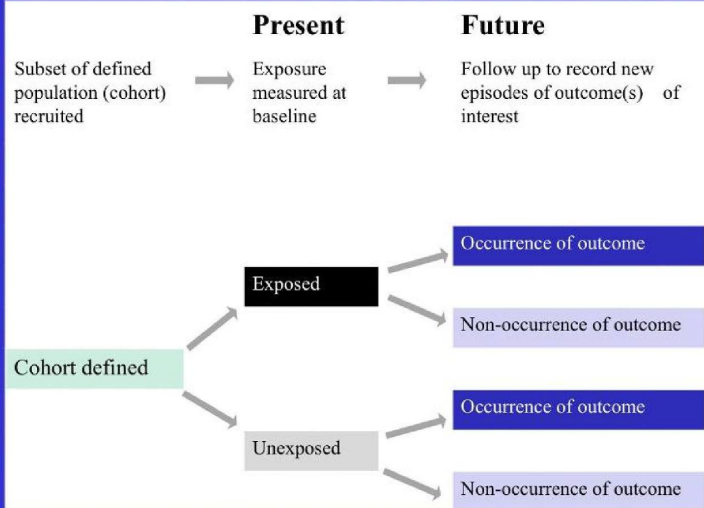
Sort individuals into equal-sized groups (block) of similar individuals, with a group size = # treatments

Randomly assign a treatment to each subject in such a way that each group gets all the treatments, one per individual, and each treatment goes to some individual in every group

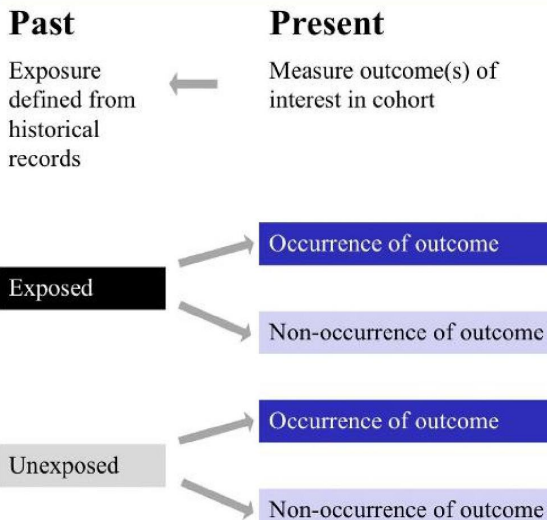
Experimental unit: each individual

1) Cohort Studies

Prospective Cohort Study



Historical Cohort Study



Selection Bias in Cohort Studies

Selection of exposed and unexposed participants relates to the outcome of interest

Prospective cohort study: participants (exposed and unexposed) selected before the outcome has occurred. Selection bias unlikely.

Historical cohort study: Exposure and outcome have already occurred when participants are selected. Selection of participants may be related to developing the outcome of interest.

Loss to follow-up Bias in Cohort Studies

Difference in retention of participants during the follow-up period.

If loss of participants occurs equally from exposed and unexposed groups (non-differentially) then the estimate of the association between exposure and outcome will be unbiased.

Loss to follow-up bias may occur if loss to follow up occurs differentially in exposed and unexposed and the reason for loss to follow-up is related to the outcome.

Measurement Bias in Cohort Studies

- Measurement error leading to subsequent misclassification of exposure and/or outcome
- Non-differential misclassification
 - errors occur equally across groups leading to underestimate of association (weakening of true association)
- Differential misclassification
 - errors occurring differently across groups leads to over- or underestimate of association.

Cohort Studies

Advantages

Clearer temporal relationship: exposure precedes disease onset
Can test hypotheses about more than one disease outcome

Disadvantages

Time consuming

Expensive

Rare diseases require very large sample sizes

Loss to follow up may introduce bias

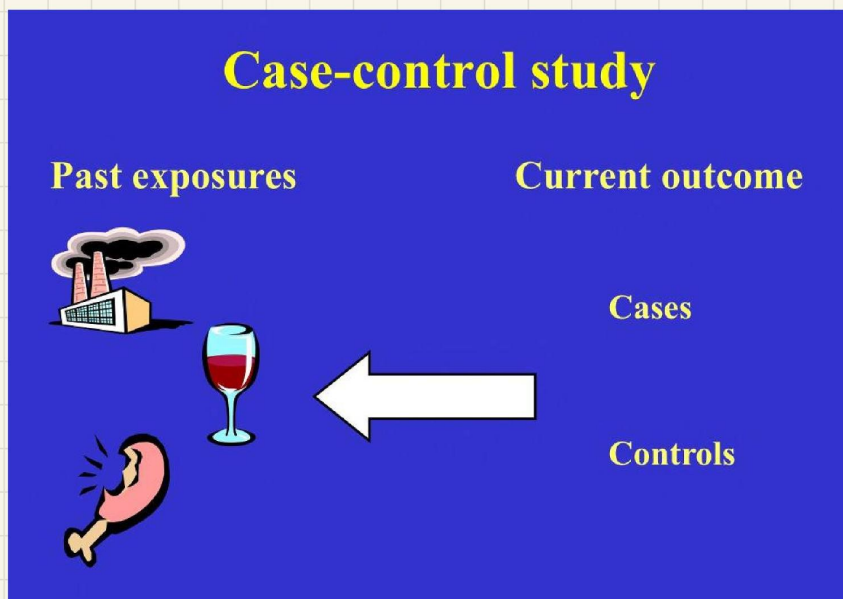
Changes over time can affect exposure and disease classification

2) Case-control studies

Measure the frequency of an exposure, amongst cases (with disease) and control group (without disease)
Start with subjects who either have or have not got disease and look retrospectively for possible exposures.

If exposure is more common amongst cases, it is associated with an increased risk of disease

If it is less common amongst cases, it may be a protective factor.



Intuitively obvious and deceptively simple design - extension of clinical history
Measure the frequency of an exposure, amongst cases (with disease) and control group (without disease or with another disease)

Assess exposure retrospectively

If exposure is more common among cases, it is associated with an increased risk of disease

If it is less common amongst cases, it may be a protective factor.

Case Definition

Pathological - histological diagnosis of breast cancer

Radiological - CT scan of a brain tumour

Clinical - Parkinson's disease diagnosed by neurologist

Self-Report - GHQ depression

Principle of Control Selection

- Controls must be individuals who would have been designated study cases if they had developed the disease in question
- Cohort analogy - the controls are a representative sample of individuals who are non-diseased at the end of follow-up
- Selection bias occurs when the controls are not a representative sample from the at-risk population
 - cases are patients with multiple sclerosis
 - exposure is dietary fat consumption
 - controls are members of staff working for the neurologist

Potential Controls

- Other disease controls
 - Hospital controls
 - Cancer registry controls
- "Healthy" controls
 - Primary care controls
 - Spouse or family controls
 - Random digit dialling
 - Electoral register

Bias in Case Control Studies

- Selection bias
 - systematic bias in selection of participants that occurs differentially across cases and controls
 - cases and controls must be selected irrespective of their exposure status
- Loss to follow-up Bias
 - not relevant in case control studies

Measurement bias in case control studies

Differential misclassification

Recall bias: errors in recall of past exposure differ across cases and controls

Interviewer bias: interviewer enquires about past exposure differentially across cases and controls

Strengths and Weaknesses of Case Control Studies

Strengths

- relatively quick
- more efficient for rare diseases and long latency diseases
- allows examination of multiple exposures
- useful for acute exposures
- likely to be of increasing importance in genetic epidemiology

Weaknesses

- prone to recall bias
- difficulties with controls (possibility of selection bias)
- inefficient for rare exposures
- cannot calculate incidence rates
- temporal relationship may not be clear

Ecological Study

Study in which units of analysis are populations rather than individuals



Interpreting Ecological Studies

Ecological 'fallacy': the assumption that average characteristics for populations apply to individuals. Although on average some populations consume more salt, is it individuals who consume most who develop strokes?

Are associations causal, or is there something related to salt intake that leads to stroke? (confounding)